



Fedora Scientific

Open Source Scientific Computing

Fedora Scientific Spin is a Fedora Linux spin that aims to showcase the open source tools for scientific and numerical computing. It was first released officially with Fedora 16 in November 2011, and is targeted at current and future Linux users in the domain of scientific computing.

I started work on this spin with a simple goal, which I will put into perspective here. My work and play involves programming, writing articles, visualisation and analysis of numerical data—and hence, I use a lot of open source libraries and tools to assist me. If you haven't guessed already, I end up installing these tools every single time I do a fresh install of Linux. This motivated the need for a Linux distribution with these tools pre-installed, specifically targeted at users like me, who use Linux for scientific work. The Fedora community's popularity and the insanely simple process of creating spins provided an ideal launch pad.

Now let us take a brief look at the applications and libraries currently available in Fedora Scientific.

Applications in Fedora Scientific

The current set of applications shipped in Fedora Scientific are broadly classified into the following categories:

- **Scientific computing tools and environments:**
The numerical computing package *GNU Octave*,

the computer algebra system *Maxima*, with its front-end *wxMaxima*, the Python scientific libraries *SciPy*, *NumPy* and *Spyder* (a Python environment for scientific computing) are some of the software included in this category. A development environment for R, the statistical computing environment, is also included, and so are the *ROOT* tools for analysing large amounts of data.

- **Generic programming and development tools and libraries:** Software in this category includes the *GNU C/C++* and *FORTRAN* compilers, the *OpenJDK* Java development tools, and the IDEs *NetBeans* and *Eclipse*. Also included are *autotools*, *flex*, *bison*, *ddd* and *valgrind*.
- **Parallel and distributed programming tools/libraries:** Software tools and libraries included in this category include the popular parallel programming libraries *OpenMPI*, *PVM*, and the shared-memory programming library *OpenMP*. Also included is the